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Deluge Scripting in Zoho Creator

1. Aim

To learn the Deluge scripting language of Zoho Creator by writing and running ten standard stand-alone programs that use variables, conditions, loops, lists, maps, string functions and date functions.

2. Introduction

2.1 Deluge

Deluge (Data Enriched Language for the Universal Expression) is the scripting language built into Zoho Creator. It adds logic to an application without a separate programming environment. A Deluge script is written in the script editor of a workflow, a function, a button, a schedule or an approval. The result of a script can be shown with the info statement, which is useful while learning and testing.

2.2 Where Deluge Scripts Are Used

Deluge scripts run in various places within a Zoho Creator application, including form workflows, buttons, functions, schedules, approvals, and blueprints. The programs executed in this experiment are stand-alone programs that do not require an active form configuration and can be tested directly within the environment.

3. Objectives of the Study

• To declare variables and use arithmetic operators and the basic data types of Deluge.

• To control the flow of a script with if / else conditions and for each loops.

• To store data in lists and maps and to process text and dates with built-in functions.

• To solve common problems such as finding the largest number, counting even and odd numbers and calculating a factorial.

4. Procedure

1. Open Zoho Creator and open a practice application in Edit mode, then click Workflow in the top bar.

2. Open the Functions section and click + New Function to create a standalone function for each program, for example program_01_variables.

3. Type or paste the Deluge script of the program in the script editor.

4. Click Save, then click Execute (Run) to run the function.

5. Read the output of the info statements in the output window of the editor.

6. Take a screenshot of the editor with the output for every program.

5. Detailed Analysis

Table 1: Deluge programs written in this experiment

No.	Program	Deluge features used
1	Variables and arithmetic operators	Variables, arithmetic, %
2	if / else if condition	if, else if, else
3	for each loop: sum and average	List, for each, size
4	List operations	add, get, contains, sort, removeElement
5	Map operations	Map, put, get, containKey, remove, keys
6	String functions	length, toUpperCase, subString, indexOf, replaceAll, toList
7	Date functions	zoho.currentdate, addDay, addMonth, toString
8	Largest and smallest in a list	List, for each, if, comparison
9	Counting even and odd numbers	for each, if / else, % operator
10	Multiplication table and factorial	for each, info, multiplication

Table 2: Places where Deluge scripts are used

Place	Use of the Deluge script
Form workflow	Fills fields, validates data and updates other records when a record is created or edited
Button	Runs a script when the user clicks a button on a form, report or page
Function	A reusable script that is called from workflows, buttons and schedules
Schedule	Runs a script at a fixed date and time
Approval	Runs a script when a record is approved or rejected
Blueprint	Runs a script when a record moves from one stage to the next

6. Programs, Scripts, and Sample Outputs

Program 1: Variables and arithmetic operators

Deluge Script

```
price = 250;
quantity = 4;
discount = 10;
subtotal = price * quantity;
discountAmt = subtotal * discount / 100;
total = subtotal - discountAmt;
info "Subtotal : " + subtotal;
info "Discount : " + discountAmt;
info "Total : " + total;
info "Remainder : " + (subtotal % 3);
```

Sample Output

```
Subtotal : 1000
Discount : 100.0
Total : 900.0
Remainder : 1
```

Program 2: if / else if condition

Deluge Script

```
marks = 72;
if(marks >= 90)
{
 grade = "A+";
}
else if(marks >= 75)
{
 grade = "A";
}
else if(marks >= 60)
{
 grade = "B";
}
else
{
 grade = "C";
}
info "Marks : " + marks;
info "Grade : " + grade;
```

Sample Output

```
Marks : 72
Grade : B
```

Program 3: for each loop: sum and average

Deluge Script

```
marks = List();
marks.add(78);
marks.add(85);
marks.add(92);
marks.add(66);
sum = 0;
for each m in marks
{
 sum = sum + m;
}
average = sum / marks.size();
info "Sum : " + sum;
info "Average : " + average;
```

Sample Output

```
Sum : 321
Average : 80.25
```

Program 4: List operations

Deluge Script

```
names = List();
names.add("Ravi");
names.add("Anita");
names.add("Meena");
info "Size : " + names.size();
info "Second : " + names.get(1);
info "Has Anita : " + names.contains("Anita"
);
info "Sorted : " + names.sort(true);
names.removeElement("Ravi");
info "After remove : " + names;
```

Sample Output

```
Size : 3
Second : Anita
Has Anita : true
Sorted : [Anita, Meena, Ravi]
After remove : [Anita, Meena]
```

Program 5: Map operations

Deluge Script

```
student = Map();
student.put("Name", "Anita");
student.put("Dept", "CSE");
student.put("Marks", 88);
info student.get("Name");
info student.containKey("Dept");
student.remove("Dept");
info student.keys();
info student.size();
```

Sample Output

```
Anita
true
[Name, Marks]
2
```

Program 6: String functions

Deluge Script

```
text = "Zoho Creator Lab";
info text.length();
info text.toUpperCase();
info text.toLowerCase();
info text.subString(5, 12);
info text.indexOf("Creator");
info text.replaceAll("Lab", "Record");
info text.contains("Zoho");
words = text.toList(" ");
info words.size();
```

Sample Output

```
16
ZOHO CREATOR LAB
zoho creator lab
Creator
5
Zoho Creator Record
true
3
```

Program 7: Date functions

Deluge Script

```
today = zoho.currentdate;
info "Today : " + today;
info "Next week : " + today.addDay(7);
info "Next month : " + today.addMonth(1);
info "Day name : " + today.toString("EEEE");
info "Formatted : " + today.toString("dd/MM/
yyyy");
info "Year : " + today.getYear();
```

Sample Output

```
Today : 18-Oct-2026
Next week : 25-Oct-2026
Next month : 18-Nov-2026
Day name : Sunday
Formatted : 18/10/2026
Year : 2026
```

Program 8: Largest and smallest in a list

Deluge Script

```
nums = List();
nums.add(45);
nums.add(12);
nums.add(78);
nums.add(33);
largest = nums.get(0);
smallest = nums.get(0);
for each n in nums
{
 if(n > largest)
 {
 largest = n;
 }
 if(n < smallest)
 {
 smallest = n;
 }
}
info "Largest : " + largest;
info "Smallest : " + smallest;
```

Sample Output

```
Largest : 78
Smallest : 12
```

Program 9: Counting even and odd numbers

Deluge Script

```
nums = List();
nums.add(3);
nums.add(8);
nums.add(15);
nums.add(20);
nums.add(7);
evenCount = 0;
oddCount = 0;
for each n in nums
{
 if(n % 2 == 0)
 {
 evenCount = evenCount + 1;
 }
 else
 {
 oddCount = oddCount + 1;
 }
}
info "Even numbers : " + evenCount;
info "Odd numbers : " + oddCount;
```

Sample Output

```
Even numbers : 2
Odd numbers : 3
```

Program 10: Multiplication table and factorial

Deluge Script

Sample Output

```
nums = List();
nums.add(1);
nums.add(2);
nums.add(3);
nums.add(4);
nums.add(5);
fact = 1;
for each n in nums
{
 fact = fact * n;
 info "3 x " + n + " = " + (3 * n);
}
info "Factorial of 5 : " + fact;
```

```
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
Factorial of 5 : 120
```

7. Result

Ten standard Deluge programs were successfully written and run in Zoho Creator. The scripts covered variables, conditional elements, loops, lists, maps, string operations, date functions, and common numerical tasks. Each execution yielded the expected outputs, demonstrating the utility of Deluge scripting as the foundational logic block for building dynamic workflow features within Zoho Creator applications.